

```
mysql> create database company_db;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use company_db;
Database changed
mysql> CREATE TABLE roles(role_id INT PRIMARY KEY, role_name VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE users(user_id INT PRIMARY KEY,name VARCHAR(50),email VARCHAR(50),role_id INT);
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE departments(dept_id INT PRIMARY KEY,dept_name VARCHAR(50));
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE employees(emp_id INT PRIMARY KEY,emp_name VARCHAR(50),dept_id INT,manager_id INT,salary DECIMAL(10,2));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE customers(customer_id INT PRIMARY KEY,customer_name VARCHAR(50),city VARCHAR(50));
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE orders(order_id INT PRIMARY KEY,customer_id INT,order_date DATE,total_amount DECIMAL(10,2));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE accounts(account_no INT PRIMARY KEY,holder_name VARCHAR(50),balance DECIMAL(10,2));
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> INSERT INTO roles (role_id, role_name) VALUES
-> (1, 'Admin'),
-> (2, 'Manager'),
-> (3, 'Employee'),
-> (4, 'HR'),
-> (5, 'Intern');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO users (user_id, name, email, role_id) VALUES
-> (101, 'Rahul', 'rahul@gmail.com', 1),
-> (102, 'Priya', 'priya@gmail.com', 2),
-> (103, 'Amit', 'amit@gmail.com', 3),
-> (104, 'Sneha', 'sneha@gmail.com', 4),
-> (105, 'Kiran', 'kiran@gmail.com', 5);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO departments (dept_id, dept_name) VALUES
-> (10, 'HR'),
-> (20, 'Finance'),
-> (30, 'IT'),
-> (40, 'Marketing'),
-> (50, 'Sales');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO employees (emp_id, emp_name, dept_id, manager_id, salary) VALUES
-> (201, 'Ramesh', 30, NULL, 90000.00),
-> (202, 'Suresh', 30, 201, 60000.00),
-> (203, 'Anita', 20, 201, 55000.00),
-> (204, 'Pooja', 10, 201, 50000.00),
-> (205, 'Vijay', 50, 202, 45000.00);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO customers (customer_id, customer_name, city) VALUES
-> (301, 'Arjun', 'Hyderabad'),
-> (302, 'Meena', 'Chennai'),
-> (303, 'Ravi', 'Bangalore'),
-> (304, 'Neha', 'Mumbai'),
-> (305, 'Karthik', 'Delhi');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO orders (order_id, customer_id, order_date, total_amount) VALUES
-> (401, 301, '2026-08-01', 2500.00),
-> (402, 302, '2026-08-02', 4200.50),
-> (403, 303, '2026-08-02', 1800.75),
-> (404, 301, '2026-08-03', 3500.00),
-> (405, 305, '2026-08-04', 6200.25);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO accounts (account_no, holder_name, balance) VALUES
-> (1001, 'Rahul', 50000.00),
-> (1002, 'Priya', 75000.50),
-> (1003, 'Amit', 32000.75),
-> (1004, 'Sneha', 98000.00),
-> (1005, 'Kiran', 15000.25);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT emp_name,dept_name FROM employees e INNER JOIN departments d ON e.dept_id=d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Ramesh   | IT        |
| Suresh   | IT        |
| Anita    | Finance   |
| Pooja    | HR        |
| Vijay    | Sales     |
+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+-----+
| name | email | role_name |
+-----+-----+-----+
| Rahul | rahul@gmail.com | Admin |
| Priya | priya@gmail.com | Manager |
| Amit | amit@gmail.com | Employee |
| Sneha | sneha@gmail.com | HR |
| Kiran | kiran@gmail.com | Intern |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT name,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+
| name | role_name |
+-----+-----+
| Rahul | Admin |
| Priya | Manager |
| Amit | Employee |
| Sneha | HR |
| Kiran | Intern |
+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id UNION SELECT name,email,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+-----+
| name | email | role_name |
+-----+-----+-----+
| Rahul | rahul@gmail.com | Admin |
| Priya | priya@gmail.com | Manager |
| Amit | amit@gmail.com | Employee |
| Sneha | sneha@gmail.com | HR |
| Kiran | kiran@gmail.com | Intern |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT e.emp_name Employee,m.emp_name Manager FROM employees e LEFT JOIN employees m ON e.manager_id=m.
emp_id;
+-----+-----+
| Employee | Manager |
+-----+-----+
| Ramesh   | NULL    |
| Suresh   | Ramesh  |
| Anita    | Ramesh  |
| Pooja    | Ramesh  |
| Vijay    | Suresh  |
+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select o.order_id, customer_name, city, total_amount from customers c join orders o on c.customer_id =
o.customer_id;
+-----+-----+-----+-----+
| order_id | customer_name | city | total_amount |
+-----+-----+-----+-----+
| 401      | Arjun         | Hyderabad | 2500.00 |
| 402      | Meena        | Chennai  | 4200.50 |
| 403      | Ravi         | Bangalore | 1800.75 |
| 404      | Arjun        | Hyderabad | 3500.00 |
| 405      | Karthik      | Delhi    | 6200.25 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT dept_name,COUNT(*),AVG(salary),MAX(salary) FROM employees e JOIN departments d ON e.dept_id=d.de
pt_id GROUP BY dept_name;
+-----+-----+-----+-----+
| dept_name | COUNT(*) | AVG(salary) | MAX(salary) |
+-----+-----+-----+-----+
| IT        | 2        | 75000.000000 | 90000.00    |
| Finance   | 1        | 55000.000000 | 55000.00    |
| HR        | 1        | 50000.000000 | 50000.00    |
| Sales     | 1        | 45000.000000 | 45000.00    |
+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

```
mysql> SELECT customer_name,SUM(total_amount) TotalPurchase FROM customers c JOIN orders o ON c.customer_id=o.
customer_id GROUP BY customer_name ORDER BY TotalPurchase DESC;
+-----+-----+
| customer_name | TotalPurchase |
+-----+-----+
| Karthik      | 6200.25      |
| Arjun        | 6000.00      |
| Meena        | 4200.50      |
| Ravi         | 1800.75      |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT student_name,course_name,faculty_name FROM students s JOIN enrollments e ON s.student_id=e.stude
nt_id JOIN courses c ON e.course_id=c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;
ERROR 1146 (42S02): Table 'company_db.students' doesn't exist
mysql> START TRANSACTION; INSERT INTO accounts VALUES(104,'Anil',25000); COMMIT;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

```
mysql> START TRANSACTION; INSERT INTO accounts VALUES(105,'Ravi',10000); ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.02 sec)
```

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance-5000 WHERE account_no=101; UPDATE accounts SET b  
alance=balance+5000 WHERE account_no=102; COMMIT;  
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.01 sec)  
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)  
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance-3000 WHERE account_no=101; UPDATE accounts SET b  
alance=balance+3000 WHERE account_no=999; ROLLBACK;  
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)  
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)  
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance+2000 WHERE account_no=101; SAVEPOINT s1; UPDATE  
accounts SET balance=balance-1000 WHERE account_no=101; ROLLBACK TO s1; COMMIT;  
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)  
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)  
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```